COURSE PLAN

Department :	**Electronics and Communication**	
Course Name & code :	**Electromagnetic Waves**	**ECE - 2125**
Semester & branch :	**III Semester**	**Electronics and Communication**
Name of the faculty :	**1. Dr. Sampath Kumar** **2. Dr. Tanweer** **3. Dr. Karthi Pradeep** **4. Dr. Atanu Das**	

No of contact hours/week:

L	T	P	C
3	0	0	3

COURSE OUTCOMES

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	PSO	BL
CO1	Apply the governing laws of electromagnetic theory	12	30	1,2,3	2	2
CO2	Apply Maxwell's equations to derive wave equation and solve the same in simple lossy as well as lossless geometrical structures	4	20	1,2,3	2	3
CO3	Compute transmission and reflection coefficients for normal and oblique incidence of TEM waves.	4	10	1,2,3	2	3
CO4	Use transmission line parameters and analyze propagation in transmission lines	8	25	1,2,3	2	3
CO5	Analyze guided wave propagation through infinite conducting planes and rectangular waveguides	8	15	1,2,3	2	4
	Total	**36**	**100**			

COURSE LEARNING OUTCOMES (CLOS)

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes(POs)	LO	BL
CLO1	Apply the governing laws of electromagnetic theory	12	30	1,2,3	1,2,3,4,18	2,3,4
CLO2	Apply Maxwell's equations to derive wave equation and solve the same in simple lossy as well as lossless geometrical structures	4	20	1,2,3	1,2,4,18	2,3,4
CLO3	Estimate transmission and reflection coefficients for normal and oblique incidence of TEM waves.	4	10	1,2,3	1,2,4,18	3
CLO4	Use transmission line parameters and analyze propagation in transmission lines	8	25	1,2,3	1,2,3,4,18	3
CLO5	Analyze guided wave propagation through infinite conducting planes and rectangular waveguides	8	15	1,2,3	1,2,4,18	3
	Total	**36**	**100**			

IN – SEMESTER ASSESSMENTS								
S. No.	**Assessment Mode**	**Assessment Method**	**Time Duration**	**Marks**	**Weightage**	**Typology of Questions (Recommended)**	**Schedule**	**Topics Covered**
1	**MISAC 1**	**Quiz 1**	**15 Mins**	**5**	10 MCQs × 0.5 Marks = 5 marks	Bloom's taxonomy (B) level of the question should be L2and above.		L1 - L10
	MISAC 2	**Mid-Term Examination**	**60 Mins**	**30**	Descriptive: 30 M	Bloom's taxonomy (BT) level of the question should be L2 and above.		L1 – L20
	MISAC 3	**Quiz 2**	**15 Mins**	**5**	10 MCQs × ½ = 5 marks	Bloom's taxonomy (BT) level of the question should be L2 and above.		L21 – L30
	FISAC	**Carry home Assignment**	**-**	**10**	10 questions of 1 marks each	Bloom's taxonomy (BT) level of the question should be L3 and above.		L31 - L36
END – SEMESTER ASSESSMENT								
1	**Regular/Make–Up Exam**		180 Mins	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5/6 marks.	Bloom's taxonomy (BT) level of the question should be L3 and above.	17th week of the semester	Comprehensive examination covering full syllabus.

LESSON PLAN

L. No./ T. No.	Topics	Course Outcome Addressed
L0	Introduction	-
L1	Review of Vector analysis, Cartesian co-ordinate systems	CO1
L2	Cylindrical co-ordinate systems & Conversion from one to other	CO1
L3	Spherical co-ordinate systems & Conversion from one to other	CO1
T1	Tutorial	CO1
L4	Electric flux and electric flux density	CO1
L5	Gauss' law and applications	CO1
L6	Divergence and Divergence theorem	CO1
T2	Tutorial	CO1
L7	Energy, Force, Potential difference, Potential difference due to point charge	CO1
L8	Potential difference due to ring charge, potential gradient	CO1
L9	Dielectrics, Boundary conditions, Relaxation time	CO1
T3	Tutorial	CO1
L10	Magnetic field intensity, Biot-Savart's law, Ampere's law	CO1
L11	Stoke's Theorem, Magnetic Flux, Flux density, Boudary conditions	CO1
L12	Faraday's law, Displacement current, Energy densities	CO1
T4	Tutorial	CO1
L13	Maxwell's equations in integral and point form for free space and material media, Maxwell's equations in sinusoidal form	CO2
L14	Wave equations and its solutions	CO2
L15	Uniform plane wave, Wave propagation in free space, dieelctrics	CO2
T5	Tutorial	CO2
L16	Poynting vector and complex Poynting vector, Skin effect	CO2
L17	Wave polarization, Normal incidence, Transmission and reflection coefficients	CO2
L18	Standing wave ratio, Multiple interfaces	CO2
T6	Tutorial	CO2

L19	Propagation in general directions, oblique incidence, Total internal reflection and Brewster angle	CO2
L20	Parameters of transmission line at radio frequencies-lumped and distributed parameters	CO3
L21	Transmission line equation, Characteristic impedance, Reflection coefficient	CO3
T7	Tutorial	CO3
L22	Standing waves on transmission line for different loads	CO3
L23	Power and impedance relations	CO3
L24	Impedance matching	CO3
T8	Tutorial	CO3
L25	Eight, quarter and half wave lines	CO3
L26	Smith chart, construction	CO3
L27	Single stub matching	CO3
T9	Tutorial	CO3
L28	Double stub matching	CO3
L29	TE waves in infinite parallel conducting planes	CO5
L30	TM and TE waves in infinite parallel conducting planes	CO5
T10	Tutorial	CO5
L31	Characteristics of TE and TM waves, wave impedance	CO5
L32	Attenuation in guided waves	CO5
L33	Rectangular waveguides	CO5
T11	Tutorial	CO5
L34	TE waves in rectangular waveguides	CO5
L35	TM waves in rectangular waveguides	CO5
L36	Characteristics of TE and TM mode and excitation of waveguides	CO5
T12	Tutorial	CO5
L/T		

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	2	2											2	
CO2	2	2	2											2	
CO3	2	2	2											2	
CO4	2	2	2											2	
CO5	2	2	2											2	
Articulation Level	2	2	2											2	

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION
Dr. Sampath Kumar	A
Dr. Atanu Das	B
Dr. Karthi Pradeep	C
Dr. Tanweer	D
Dr. Atanu Das	VLSI

References:

Submitted by: Dr. Sampath Kumar

(Signature of the faculty)

Date: 22/07/2025

Approved by: Dr. Pallavi R Mane

(Signature of HOD)